

Henry Matar

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Education

University of Colorado Boulder, College of Engineering and Applied Science

Expected May 2027

B.S. in Electrical Engineering

Minors in Computer Science & Applied Mathematics

- **GPA: 3.98/4.00** | Dean's list: All Eligible Semesters
- Tau Beta Pi | Engineering Honors Society

Skills

Software: Altium Designer, Linux, C, C++, Java, Git, GitHub, GitLab, JIRA, Verilog, MATLAB, Python, MicroPython, SPICE (SIMetrix, LTSpice), STM32, CAD (Onshape), Power BI

Hardware: PCB Design, Embedded Systems, Oscilloscopes, Soldering, FPGA Development, Prototyping, 3D Printing

Work Experience

BAE Systems / CU Boulder Capstone Design Project – *Technical Lead, Post-Processing*

August 2026 – Present

- Developing a bistatic drone-based RF radar system with 2D imaging, object localization, and scattering characterization.

Lockheed Martin, Systems Engineering Intern – Mount Laurel, NJ

June 2026 – August 2026

- Automated end-to-end messaging validation by scripting REST API-driven tests against message topics, enabling repeatable integration and L2 testing across team products.
- Developed and integrated a Java/Spring Boot aggregator service with ActiveMQ and Hermes (Node/Angular) over AMQP 1.0, supporting reliable end-to-end message flows.
- Engineered a comprehensive JUnit 5/Mockito test suite for a Spring Boot-based tactical track manager, validating partial state merges and Protobuf-to-POJO translations across Kafka and ActiveMQ transport layers.
- Designed deep-nesting test cases for ECEF coordinates and covariance matrices to prevent null-overwriting of mission-critical track data, using Mockito static mocking and Spring's ReflectionTestUtils to isolate external dependencies.
- Refactored a Spring translation service into a modular architecture (Air, Land, Surface) and replaced an unordered HashMap with a TreeMap for relay tracking, guaranteeing deterministic ordering for IBCS aggregation and Protobuf-based health reports.

Lower Colorado River Authority (LCRA), Corporate Strategy Intern - Austin, TX

June 2025 - August 2025

- Built an interactive Power BI dashboard tracking 4 emerging technologies (SMRs, hydrogen blending, geothermal, and carbon capture), used by a 10+ person strategy team to identify key market opportunities of next-generation energy sources.

Technical Projects

4-Layer PCB – Thevenin Resistance Instrument

Jan 2026 – May 2026

- Designed and assembled a 4-layer Arduino Uno Shield in Altium with a DAC-controlled electronic load, OPAMP/MOSFET current sink, ADC voltage sensing, I2C communication, and 1:3 input divider for voltage source characterization.
- Implemented 20 pulsed current load steps at 10% duty cycle to measure voltage drop and calculate Thevenin resistance across a 12-251 mA load range.
- Developed PCB layout across two signal layers to eliminate ground-plane cross-unders, using distributed decoupling and extensive test points to improve signal integrity and board bring-up.

PCB Design – Golden Arduino

Jan 2026 – May 2026

- Designed and assembled a 2-layer ATmega328P Arduino-compatible PCB with CH340G USB-UART, ICSP programming, power switching, TVS protection, ferrite filtering, and distributed decoupling.
- Optimized PCB layout to minimize ground plane cross-unders and loop area, achieving only two cross-unders while maintaining accessible test points for board bring-up and signal integrity measurements.
- Debugged initial bootloader failure by probing power and crystal signals, identifying a hidden short under the ATmega328P
- Characterized switching noise and transient response against a commercial Arduino Uno, measuring 4.9 ns vs. 11.0 ns 5V rise time and ~20 mV vs. ~200 mV near-field switching noise.

Embedded Software Blackjack Game System

Aug 2025 – Dec 2025

- Developed an embedded Blackjack system in C for an STM32 microcontroller, with real-time LCD graphics, button controls, randomized card generation, dynamic balance tracking, and multi-screen UI using low-level hardware drivers.
- Implemented game-state management, recursive gameplay logic, dealer-card hiding behavior, and betting mechanics, including a custom cheat-code detection sequence using timed push-button interrupts and event-driven state transitions.

Electronics Design Lab Robot, Electrical Engineer

Jan 2025 – May 2025

- Designed a Bluetooth-controlled mobile robot utilizing an HC-05 module for wireless command transmission.
- Simulated and verified all analog subsystems in SPICE (SIMetrix), including dual DC motor models, speed sensor, compensator circuits, level-shifting comparators, RC low-pass filters, and a virtual ground reference.
- Engineered a BJT-based H-bridge to provide bidirectional control and sufficient current from a single supply.
- Programmed an interrupt-driven position control system on the Arduino Nano Every, leveraging encoder pulse feedback for closed-loop control and millimeter positional precision across both motors.